

MA201A — Ch. 12 Study Guide · Lecture 1

§12.5 Lines in Space · §12.6 Cylinders & Quadric Surfaces — Stewart 9e · Fall 2026

1 · Recap & Visual Intuition

Cylinders: the missing-variable rule

◆ Definition

A **cylinder** is a surface made of all lines (**rulings**) parallel to a given line, passing through a given plane curve. It does **not** have to be circular — it is any 2D curve extruded through space.

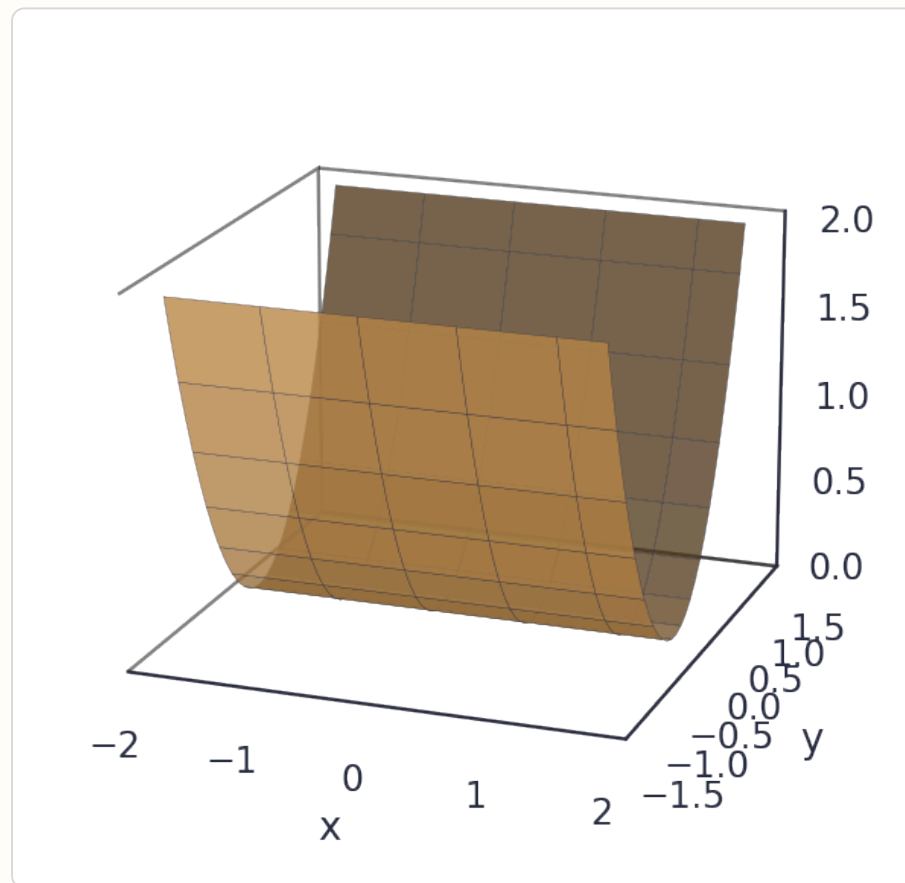
★ Key Tip

Missing-variable rule — if an equation in $\mathbb{R}^3$ lacks one coordinate, graph the curve in the plane of the two variables present, then extrude it infinitely along the missing axis, through all 8 octants.

Circular: $x^2 + y^2 = r^2 \rightarrow$ circle in xy -plane, rulings $\parallel z$

Elliptic: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a \neq b \rightarrow$ ellipse, not a circle

Parabolic: $z = y^2 \rightarrow$ parabola in yz -plane, extruded along x



$z = y^2$: a parabola in the yz -plane dragged straight out along x . Nothing constrains x , so it runs forever.

2D $\rightarrow$ 3D: parameterization carries order, direction, and speed

Segment from $(0, 1)$ to $(1, 0)$ lies on $y = 1 - x$. Let $x(t) = t \Rightarrow y(t) = 1 - t$:

$$\vec{r}(t) = \langle t, 1 - t \rangle, \quad t \in [0, 1]$$

t	(x, y)	$\vec{r}(t)$	travel
0	$(0, 1)$	$\langle 0, 1 \rangle$	start
$\frac{1}{4}$	$(\frac{1}{4}, \frac{3}{4})$	$\langle \frac{1}{4}, \frac{3}{4} \rangle$	↓
$\frac{1}{2}$	$(\frac{1}{2}, \frac{1}{2})$	$\langle \frac{1}{2}, \frac{1}{2} \rangle$	↓
$\frac{3}{4}$	$(\frac{3}{4}, \frac{1}{4})$	$\langle \frac{3}{4}, \frac{1}{4} \rangle$	↓
1	$(1, 0)$	$\langle 1, 0 \rangle$	end

★ Why the table matters

The **set** of points is just the segment — but the parameterization also fixes **which end you start at, which way you travel**, and **how fast**. Equal steps in t give equal steps in space here, so the speed is constant.

Direction vector $\vec{v} = \langle 1, -1 \rangle$; base point $(0, 1)$. For the **full infinite line**, let $t \in \mathbb{R}$ instead of $[0, 1]$.

Lines in $\mathbb{R}^3$

A line L needs exactly two things: a **base point** $P_0(x_0, y_0, z_0)$ and a **nonzero direction vector** $\vec{v} = \langle a, b, c \rangle \parallel L$. The components a, b, c are the **direction numbers**.

◆ Vector equation

$$\vec{r}(t) = \vec{r}_0 + t\vec{v} = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

◆ Parametric equations

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

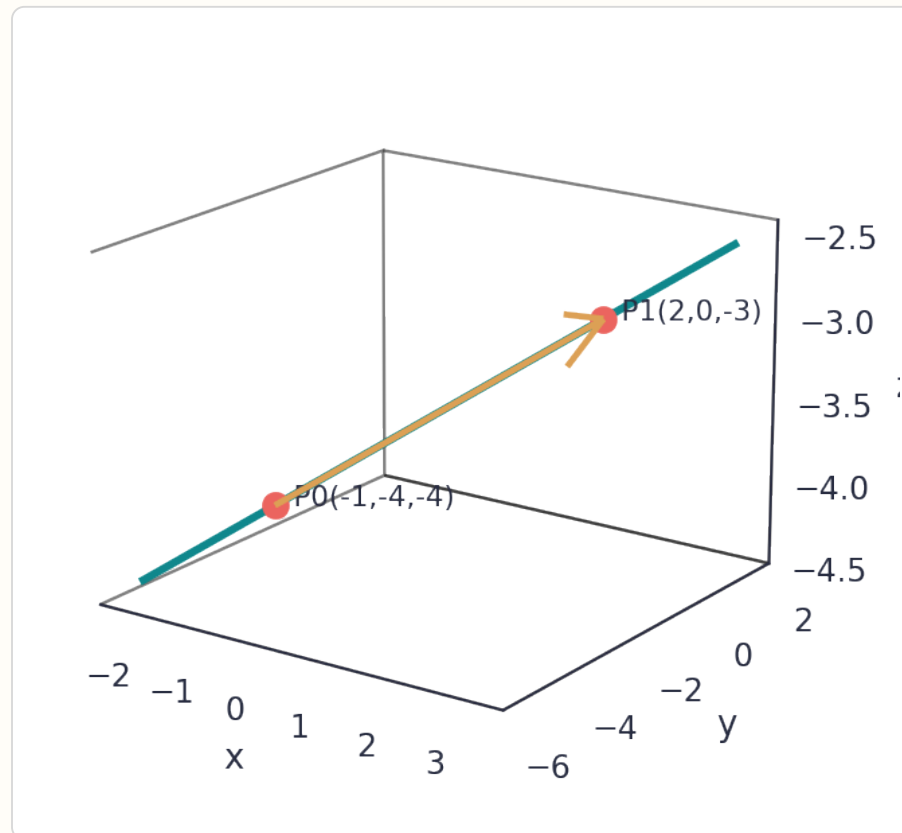
◆ Symmetric equations

(eliminate t — requires $a, b, c \neq 0$)

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

◆ Segment $P_0 \rightarrow P_1$

$$\vec{r}(t) = (1 - t)\vec{r}_0 + t\vec{r}_1, \quad 0 \leq t \leq 1$$



Line through $P_0(-1, -4, -4)$ and $P_1(2, 0, -3)$, direction $\vec{v} = \langle 3, 4, 1 \rangle$ — worked in §4.

✎ Stewart 9e §12.5 · §12.6

✎ Paul's Online Notes — Lines / Quadric Surfaces

► Prof. Leonard — Calc 3, 12.5–12.6

2 · Classification Cheat-Sheet

★ Key Tip

General second-degree form: $Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$. Fast ID on a timed question: **count the minus signs**, check the **right-hand side** ($= 1$ vs $= 0$), and look for an **isolated linear variable**.

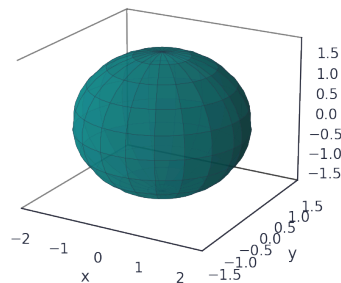
Algebraic signature	Surface	Axis / opening	Horizontal traces	Vertical traces
All +, = 1	Ellipsoid	bounded — no axis	ellipses	ellipses
Exactly one −, = 1	Hyperboloid, 1 sheet	the negative variable	ellipses	hyperbolas

Exactly two $-$, $= 1$	Hyperboloid, 2 sheets	the positive variable	ellipses ($ z \geq c$)	hyperbolas
All degree 2, $= 0$	Elliptic cone	the isolated variable	ellipses	hyperbolas / lines
One linear var, other two same sign	Elliptic paraboloid	opens along linear var	ellipses	parabolas
One linear var, other two opposite sign	Hyperbolic paraboloid	saddle at origin	hyperbolas	parabolas
A variable is missing	Cylinder	rulings $\parallel$ missing axis	the 2D curve	lines

Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

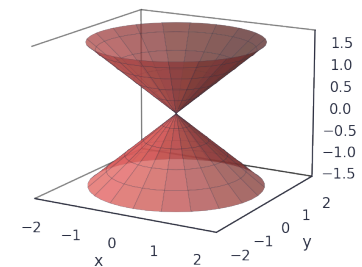
Closed, bounded. Every trace an ellipse. $a = b = c \Rightarrow$ **sphere**.



Elliptic Cone

$$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

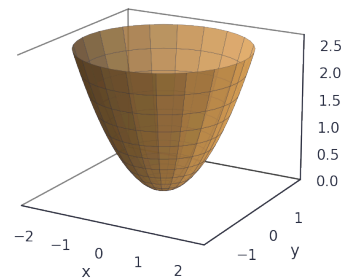
Two lobes, apex-to-apex. RHS = 0 is the giveaway.



Elliptic Paraboloid

$$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

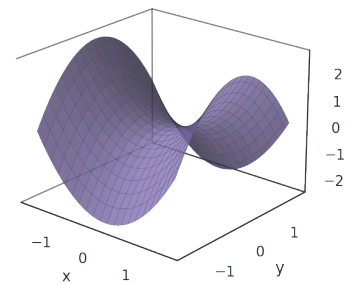
Bowl — up if $c > 0$, down if $c < 0$.



Hyperbolic Paraboloid

$$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

Saddle. Min one way, max the other.

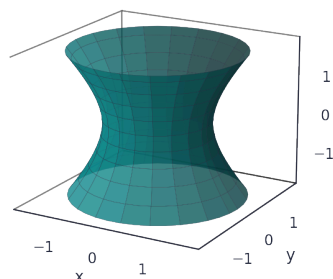


Hyperboloid — 1 Sheet

Hyperboloid — 2 Sheets

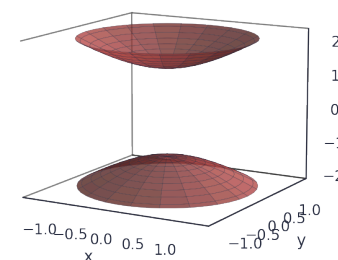
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Connected hourglass. **One** minus.



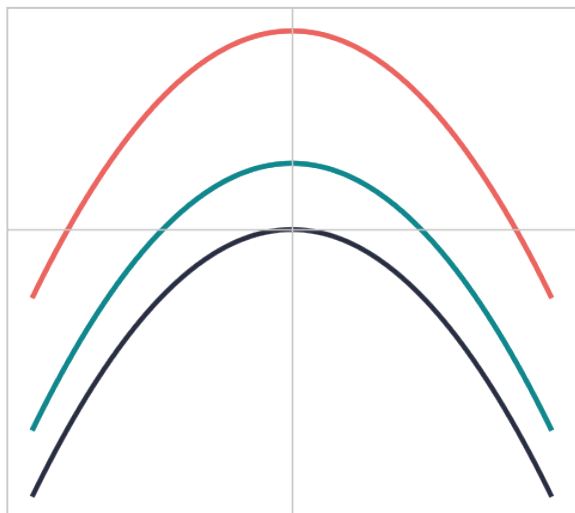
$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Two lobes apart. **Two** minuses.

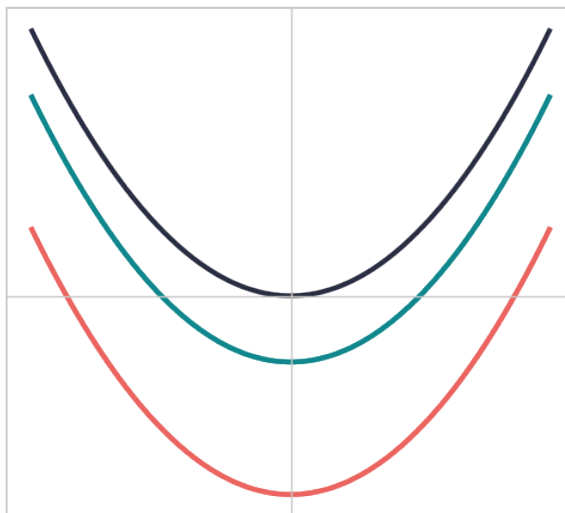


Building a saddle from its traces: $z = x^2 - y^2$

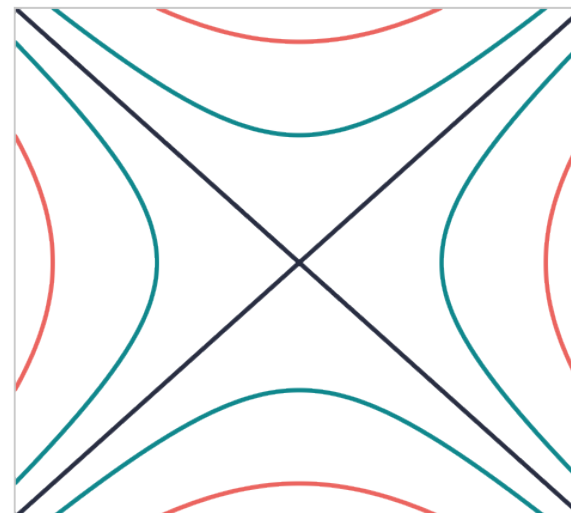
x-traces: $z = k^2 - y^2$



y-traces: $z = x^2 - k^2$



z-traces: $k = x^2 - y^2$



Slicing by constant x (downward parabolas), constant y (upward parabolas), constant z (hyperbolas, crossing lines at $k = 0$).

Synthesis

$x = k \Rightarrow z = k^2 - y^2$ (down) $\cdot$ $y = k \Rightarrow z = x^2 - k^2$ (up) $\cdot$ $z = k \Rightarrow k = x^2 - y^2$ (hyperbolas). Upward curvature along x + downward curvature along y = the saddle.

3 · Assessment Traps & Lab Hints

⚠ iClicker Alert — dimension ambiguity

$x^2 + y^2 = 1$ is a **circle** in $\mathbb{R}^2$ but a **circular cylinder** in $\mathbb{R}^3$. Always check which space the question is posed in before answering. Same trap: $y = x^2$ is a parabola in $\mathbb{R}^2$, a parabolic cylinder in $\mathbb{R}^3$.

⚠ iClicker Alert — zero direction number

Symmetric form breaks when a direction number is 0 — you cannot divide by zero. If $c = 0$, drop that fraction and state the constant coordinate on its own:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b}, \quad z = z_0$$

Geometrically the line lies in the plane $z = z_0$. Worked in full at **P2** below.

⚠ iClicker Alert — cone vs. hyperboloid

$x^2 + y^2 - z^2 = __$ — the right-hand side decides everything: $= 0 \rightarrow$ **cone**; $= 1 \rightarrow$ **hyperboloid of one sheet**; $= -1 \rightarrow$ **hyperboloid of two sheets**. Identical left sides, three different surfaces.

✓ Assignment 1 — Wed Sept 23, N1044

Multi-step vector work is marked on **shown work**, not just the final line:

- Compute $\vec{v} = \overrightarrow{P_0P_1}$ **explicitly** before writing any equation.
- State the parameter domain every time — $t \in \mathbb{R}$ for a line, $0 \leq t \leq 1$ for a segment.
- For shifted quadrics, show each **completing-the-square** step; the centre/vertex is usually worth its own mark.
- Verify a segment formula by checking $t = 0$ gives P_0 and $t = 1$ gives P_1 .

☞ Maple Lab — Thurs Sept 24, LH3068

Load the plotting package once per worksheet, then:

```
with(plots):  
# a line segment (parametric, restricted domain)  
spacecurve([-1+3*t, -4+4*t, -4+t], t=0..1, thickness=3, color=red);  
# an implicit quadric  
implicitplot3d(x^2/4 + y^2/9 + z^2 = 1, x=-3..3, y=-4..4, z=-2..2);  
# an explicit surface z = f(x,y)  
plot3d(x^2 - y^2, x=-2..2, y=-2..2, style=patchcontour);
```

Curve of intersection — solve for the shared constraint, then parameterize it. E.g. $z = x^2 + y^2$ meets $z = 4$ on the circle $x^2 + y^2 = 4$, so $x = 2 \cos t$, $y = 2 \sin t$, $z = 4$:

```
spacecurve([2*cos(t), 2*sin(t), 4], t=0..2*Pi, thickness=3);  
display([surface, curve], scaling=constrained);
```

Two habits that save marks: pick ranges that actually **contain** the interesting region, and add `scaling=constrained` so a sphere doesn't render as an ellipsoid.

4 · Fully Worked Problems

🔗 P1 · Line through two points — $P_0(-1, -4, -4)$, $P_1(2, 0, -3)$

1. Direction vector.

$$\vec{v} = \overrightarrow{P_0P_1} = \langle 2 - (-1), 0 - (-4), -3 - (-4) \rangle = \langle 3, 4, 1 \rangle$$

2. Parametric equations (base P_0 , $t \in \mathbb{R}$):

$$x(t) = -1 + 3t, \quad y(t) = -4 + 4t, \quad z(t) = -4 + t$$

3. Symmetric equations (all direction numbers nonzero, so all three fractions survive):

$$\frac{x+1}{3} = \frac{y+4}{4} = \frac{z+4}{1}$$

4. Segment form via $\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1$, $0 \leq t \leq 1$:

$$x = (1-t)(-1) + t(2) = -1 + 3t, \quad y = (1-t)(-4) + t(0) = -4 + 4t, \quad z = (1-t)(-4) + t(-3) = -4 + t$$

$$\vec{r}(t) = \langle -1 + 3t, -4 + 4t, -4 + t \rangle$$

Same components as the infinite line — only the **domain** differs. Check: $t = 0 \rightarrow P_0$, $t = 1 \rightarrow P_1$. ✓

P2 · Zero direction number

L through $P_0(3, -2, 5)$ with $\vec{v} = \langle 4, 0, -7 \rangle$.

Parametric:

$$x = 3 + 4t, \quad y = -2, \quad z = 5 - 7t$$

y is **constant** — the middle direction number is 0, so y never changes.

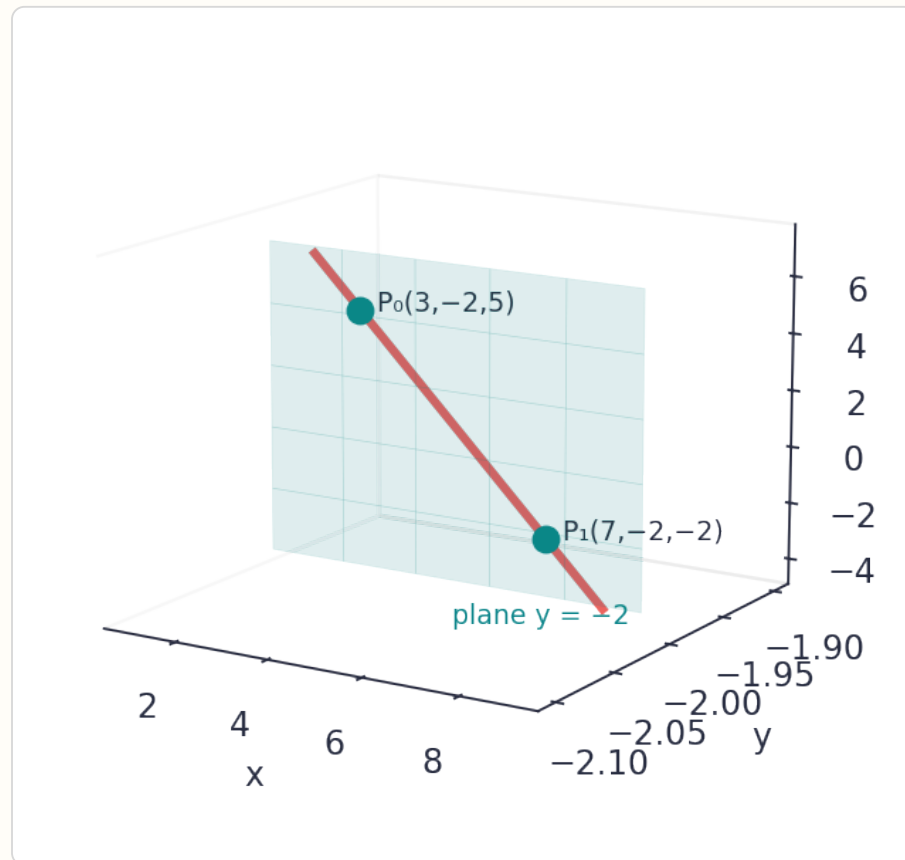
Symmetric: $b = 0$, so the y -fraction is illegal. Use the two valid fractions and state y separately:

$$\frac{x-3}{4} = \frac{z-5}{-7}, \quad y = -2$$

The line lives entirely in the plane $y = -2$.

Segment to $P_1(7, -2, -2)$: check $P_1 = P_0 + 1 \cdot \vec{v} \checkmark$, so

$$\vec{r}(t) = \langle 3 + 4t, -2, 5 - 7t \rangle, \quad 0 \leq t \leq 1$$



Because $b = 0$, the whole line is trapped in the plane $y = -2$ (shaded).

🔧 P3 · Completing the square

Classify

$$x^2 - y^2 + 4z^2 - 2x - 4y + 8z + 1 = 0$$

Group and complete each square:

$$x^2 - 2x = (x - 1)^2 - 1$$

$$-y^2 - 4y = -(y^2 + 4y) = -[(y + 2)^2 - 4] = -(y + 2)^2 + 4$$

$$4z^2 + 8z = 4(z^2 + 2z) = 4[(z + 1)^2 - 1] = 4(z + 1)^2 - 4$$

Substitute back:

$$(x - 1)^2 - 1 - (y + 2)^2 + 4 + 4(z + 1)^2 - 4 + 1 = 0$$

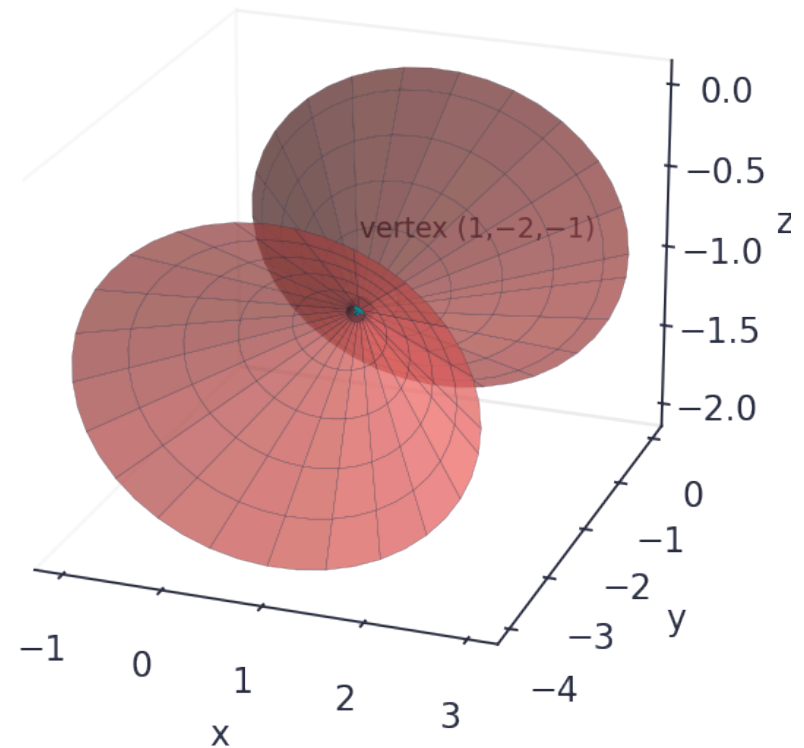
Constants: $-1 + 4 - 4 + 1 = 0$, so they cancel exactly:

$$(x - 1)^2 - (y + 2)^2 + 4(z + 1)^2 = 0$$

Identify. Right-hand side is 0 with all terms degree 2 → **elliptic cone**.
Isolating the negative variable makes the axis obvious:

$$(y + 2)^2 = (x - 1)^2 + 4(z + 1)^2$$

Axis || the y -axis; **vertex** at $(1, -2, -1)$.

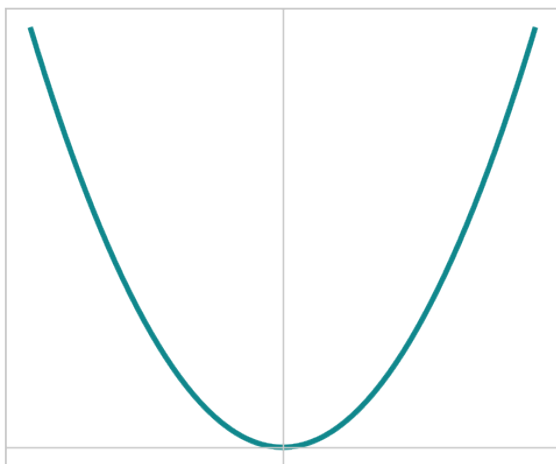


The shifted cone. Translation moves the vertex off the origin but never changes the **type**.

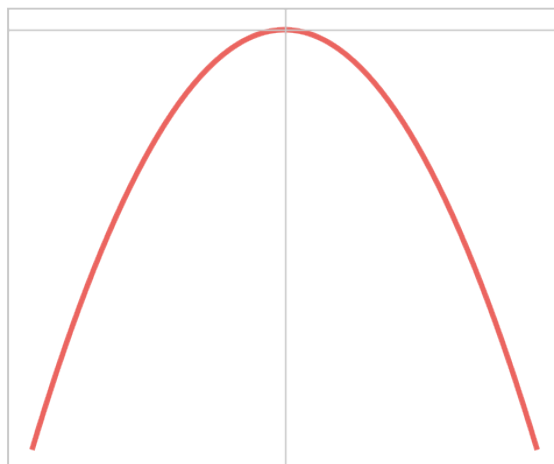
🔗 P4 · Full trace analysis — $\frac{z}{2} = \frac{y^2}{4} - \frac{x^2}{9}$

Slice	Trace equation	Curve	Reading
$x = 0$ (in yz)	$z = \frac{y^2}{2}$	upward parabola	along y , origin is a minimum
$y = 0$ (in xz)	$z = -2\frac{x^2}{9}$	downward parabola	along x , origin is a maximum
$z = 2$ ($\parallel xy$)	$\frac{y^2}{4} - \frac{x^2}{9} = 1$	hyperbola, opens along y	above the saddle
$z = -2$ ($\parallel xy$)	$\frac{x^2}{9} - \frac{y^2}{4} = 1$	hyperbola, opens along x	below — axis flips
$z = 0$ (in xy)	$y = \pm(\frac{2}{3})x$	two crossing lines	the degenerate level

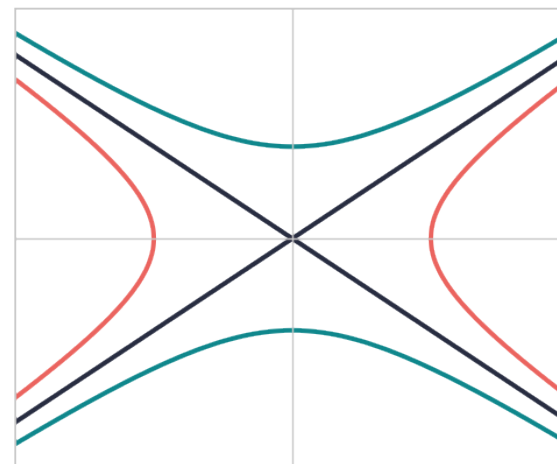
$x=0: z = y^2/2$ (min)



$y=0: z = -2x^2/9$ (max)



$z=0$ lines, $z=\pm 2$ hyperbolas



Min along y , max along x , and the level curves flipping axis as z crosses 0.

Behaviour at the origin: a saddle point — a minimum in one direction and a maximum in the perpendicular one, so it is neither a local max nor a local min. The surface is a **hyperbolic paraboloid** opening along z .

📖 Stewart 9e §12.5 1–5, 13, 17, 23–39, 45, 51–57

📖 Stewart 9e §12.6 1–7, 23–30, 41, 43, 55

5 · Test Yourself

Q1

A line passes through $A(2, -1, 6)$ and $B(2, 5, -3)$.

(a) Find $\vec{v}$ and the parametric equations.

(b) Write the symmetric equations — watch for a zero.

(c) Give the vector equation of segment $A \rightarrow B$ and verify both endpoints.

Q2

Identify each surface and state its axis (or opening direction):

(a) $9x^2 - y^2 + 4z^2 = 1$

(b) $x = 2y^2 + 3z^2$

(c) $4x^2 + y^2 - z^2 = 0$

(d) $z = x^2 - 4y^2$

Q3

Consider $x^2 + 4y^2 - 6x + 8y - z + 9 = 0$.

(a) Complete the square and write it in standard form.

(b) Name the surface and give its vertex.

(c) Write the Maple command that would plot it over a sensible window.

Answer Key

Attempt all three before reading. Skip printing this page for a clean practice copy.

A1

(a) $\vec{v} = \overrightarrow{AB} = \langle 0, 6, -9 \rangle$. Parametric: $x = 2, y = -1 + 6t, z = 6 - 9t, t \in \mathbb{R}$.

(b) $a = 0$, so the x -fraction is illegal:

$$\frac{y+1}{6} = \frac{z-6}{-9}, \quad x = 2$$

The line lies in the plane $x = 2$. (Simplifying by 3: $\frac{y+1}{2} = \frac{z-6}{-3}$ is equally valid.)

(c) $\vec{r}(t) = (1-t)\langle 2, -1, 6 \rangle + t\langle 2, 5, -3 \rangle = \langle 2, -1+6t, 6-9t \rangle, 0 \leq t \leq 1$. Check $t = 0 \rightarrow (2, -1, 6) = A \checkmark, t = 1 \rightarrow (2, 5, -3) = B \checkmark$.

A2

(a) One minus sign, RHS = 1 $\rightarrow$ **hyperboloid of one sheet**, axis = y (the negative variable).

(b) x is the isolated **linear** variable, other two coefficients share a sign $\rightarrow$ **elliptic paraboloid**, opening along $+x$.

(c) All degree 2, RHS = 0 $\rightarrow$ **elliptic cone**, axis = z (the isolated negative variable).

(d) z linear, x^2 and y^2 **opposite** signs $\rightarrow$ **hyperbolic paraboloid** (saddle), opening along z .

A3

(a) Only x and y are quadratic; z stays linear.

$$\begin{aligned} x^2 - 6x &= (x-3)^2 - 9, & 4y^2 + 8y &= 4(y+1)^2 - 4 \\ (x-3)^2 - 9 + 4(y+1)^2 - 4 - z + 9 &= 0 & \Rightarrow & (x-3)^2 + 4(y+1)^2 - 4 = z \\ z + 4 &= (x-3)^2 + 4(y+1)^2 \end{aligned}$$

(b) **Elliptic paraboloid**, opening along $+z$, with vertex $(3, -1, -4)$.

(c) Solve for z and centre the window on the vertex:

```
with(plots):  
plot3d((x-3)^2 + 4*(y+1)^2 - 4, x=0..6, y=-4..2, style=patchcontour, axes=boxed);
```